

development life cycle (SDLC) is always good.

- Once a state machine is in place, the developer can focus on tuning the behaviour of the system to meet functionality requirements and enhance business value, rather than spending time on enforcing the rules of its behaviour and other crucial non-functional attributes like code maintainability, flexibility, etc.
- When a system has its responsibilities well delegated at different states, it also ensures predictability and determinism in the system, thereby making it developer-friendly when troubleshooting.
- State machines with well-defined states and events can keep in control the growth of conditional logic as well as ad-hoc code branching out with scattered Boolean flags in the software, and thereby control the overall cyclomatic complexity.

Digressing a bit into the history of software and technology, good designs and implementations backed by state machines have always proved excellent in various verticals. These include commonly used elements in our daily life like a DBMS transaction; TCP, BGP and many other networking protocols; system processes and threads, gaming software development, etc. All these software structures have a state machine in their DNA. Even everyday occurrences like traffic lights, vending machines and other digital electronic systems could all be quickly modelled using Mealy and Moore styled finite state machines.

With all this said, let us now explore some options that the open source world has to offer developers and designers. You could enable a state machine for your code, even if it's proprietary, with the open source state machine frameworks given below, and reap the benefits of the privileges we discussed earlier.

The StateFuJ framework

StateFuJ is an open source, lightweight, Java based finite state machine (FSM) framework that could help you embed a state machine into your application. It is a useful aid when your application has to deal with many synchronous and asynchronous events and requests, while handling concurrency in a reliable way. The user has to define the state machine model (states, events and transitions) using Java annotations, and the framework auto generates the necessary FSM infrastructure for the application. You can

then define your state's actions and various transitions using the auto generated code and StateFuJ APIs. You can refer to <http://www.statefulj.org/framework/#installation> for the installation and usage steps.

SMC

State machine compiler (SMC) provides users with a framework for writing a well-defined format to define states, transitions and actions. The framework is available for multiple programming languages like C, C++, Java, Python, Scala, Objective-C, JavaScript, Ruby and VB.NET. The code that SMC generates out of the user-defined state model follows the state design pattern, with a few variations. The application then uses these generated APIs to instantiate and run the state machine. You can refer to the SMC manual (<http://smc.sourceforge.net/SmcManual.htm>) for rules to define the state model, transitions and other elements. Incorrect usage could lead to compile or runtime errors.

The spring state machine

This is an open source framework released under the Apache 2.0 licence for developers to use state machines in spring applications. It uses the builder design pattern for configuration and instantiation of states. Spring state machines provide some useful features like storing the state machine configurations in a persistent storage, UML Eclipse Papyrus modelling, Zookeeper based distributed state machine, etc. 

References

- [1] <http://www.statefulj.org/framework/>
- [2] <http://smc.sourceforge.net/>
- [3] <http://projects.spring.io/spring-statemachine/>
- [4] https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-01sc-introduction-to-electrical-engineering-and-computer-science-i-spring-2011/unit-1-software-engineering/state-machines/MIT6_01SCS11_chap04.pdf

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